COHOMOLOGICAL DESCENT FOR THE COMPARISON OF THE FPPF TOPOLOGY WITH THE SMALL ETALE TOPOS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the reduction to a single sheaf, the rewriting of a_{X,*}a_X^{-1}F, and the degeneration of the relative Leray spectral sequence with the two vanishing inputs.

Editorial note. The label lemma-cohomological-descent-etale-fppf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the five cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The fppf description of the pullback. Same chapter, the lemma describing a_{X,*}a_X^{-1} in the fppf setting through the sections of the small pullback. Supplies the identity a_{X,*}a_X^{-1}F = F used after the reduction.
- The fppf V-C statement. Same chapter, the V-C lemma in the fppf setting, giving the vanishing of the higher pushforwards of epsilon_X and the identification of its degree zero pushforward. Supplies R^i epsilon_{X,*}(a_X^{-1}F) = 0 for i > 0 and epsilon_{X,*}a_X^{-1}F = pi_X^{-1}F.
- Leray's acyclicity lemma, the relative Leray spectral sequence and the etale cohomological descent. Derived Categories, the acyclicity lemma; Cohomology on Sites, the relative Leray spectral sequence; Same chapter, the cohomological descent statement for the etale comparison. Supplies the reduction to a single sheaf, the spectral sequence used to compare, and the vanishing of the higher pushforwards of pi_X.

Source reference. Original lemma 015E: sources/stacks-chow/tags.txt, source lines 1505–1505. Prerequisite pointer invoked at proof line(s) 21838; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDS: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21725–21725. Prerequisite pointer invoked at proof line(s) 21849; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0DDI: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21308–21308. Prerequisite pointer invoked at proof line(s) 21853; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DDL: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21454–21454. Prerequisite pointer invoked at proof line(s) 21843; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0734: sources/stacks-chow/tags.txt, source lines 9048–9048. Prerequisite pointer invoked at proof line(s) 21848; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a scheme. For $K \in D^+(X_{étale})$ the map*

$$K \longrightarrow Ra_{X,*}a_X^{-1}K$$

is an isomorphism with $a_X : Sh((Sch/X)_{fppf}) \to Sh(X_{étale})$ as above.

Proof. We first reduce the statement to the case where K is given by a single abelian sheaf. Namely, represent K by a bounded below complex $\mathcal{F}^\bullet$. By the case of a sheaf we see that $\mathcal{F}^n = a_{X,*}a_X^{-1}\mathcal{F}^n$ and that the sheaves $R^q a_{X,*}a_X^{-1}\mathcal{F}^n$ are zero for $q > 0$. By Leray's acyclicity lemma (Derived Categories, Lemma 015E) applied to $a_X^{-1}\mathcal{F}^\bullet$ and the functor $a_{X,*}$ we conclude. From now on assume $K = \mathcal{F}$.

By Lemma 0DDL we have $a_{X,*}a_X^{-1}\mathcal{F} = \mathcal{F}$. Thus it suffices to show that $R^q a_{X,*}a_X^{-1}\mathcal{F} = 0$ for $q > 0$. For this we can use $a_X = \epsilon_X \circ \pi_X$ and the Leray spectral sequence (Cohomology on Sites, Lemma 0734). By Lemma 0DDS we have $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$ and $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$. By Lemma 0DDI we have $R^j\pi_{X,*}(\pi_X^{-1}\mathcal{F}) = 0$ for $j > 0$. This concludes the proof. □